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EXTENDED-ABSTRACT

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Suzume-chan: A Pet-as-a-Friend Plush Agent for Bedside Guidance and Nursing-Relevant Logging

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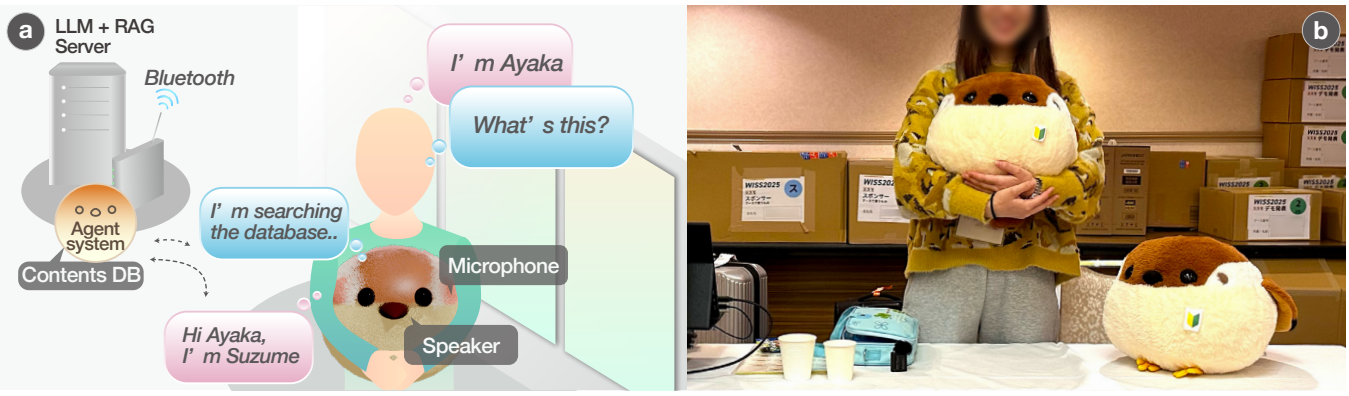

Figure 1: (a) Suzume-chan's local LLM and RAG–based architecture enabling handheld interaction, and (b) the hand-sized plush prototype demonstrated at WISS conference 2025 in Sapporo, Japan [7]

Abstract

We address a mismatch between how care information is provided and accessed. Explanations about procedures, routines, and self-management are delivered at fixed times in dense formats, leading patients to concentrate questions into nurse encounters and increasing workload. We frame this as a problem of bidirectional mediation and propose Suzume-chan, a small "Pet-as-a-Friend" plush agent that serves as an embodied information hub. Patients can speak to Suzume-chan without operating devices to receive on-demand explanations and reminders, while nurses obtain compact, nursing-relevant records. Suzume-chan runs entirely on a local network using automatic speech recognition, a local language model, retrieval-augmented generation, and text-to-speech. A workshop-style proof-of-concept highlighted embodiment, latency, and trust as key considerations for clinical use.

CCS Concepts

• **Hardware** → **Emerging interfaces**; • **Applied computing** → *Health informatics*; • **Human-centered computing** → *Empirical studies in HCI*.

Keywords

Embodied Information Agents, Bidirectional Mediation, Nursing Communication Support

*Both authors contributed equally to the paper

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1 Introduction

In nursing and medical care, patients and nurses share the same physical environment but often lack a shared understanding of relevant information. Patients are expected to learn institutional procedures, daily routines, and self-management practices at admission and during treatment. However, such information is typically delivered in dense formats and at fixed moments, when patients may have limited cognitive, emotional, or physical capacity. As a result, patients often struggle to access or retain information in a form and timing that matches their lived situations.

At the same time, nurses need ongoing insight into how patients are coping in everyday ward life—such as rest patterns, basic activities, or recurring signs of discomfort or confusion. This information is usually inferred from brief interactions and self-reports under high workload and time pressure, making systematic capture and organization difficult without increasing documentation burden.

Prior work on robot-assisted therapy has shown that soft, animal-like or pet-like robots can support emotional well-being and reduce loneliness in care settings [2]. Systems such as the therapeutic seal robot Paro demonstrate that zoomorphic, "cute" artifacts can be socially accepted, even by patients who resist overtly technological devices [6]. However, these agents primarily function as emotional companions and are rarely designed as part of the information infrastructure that organizes or mediates patient-related information for healthcare professionals.

In contrast, we focus on a plush, pet-like agent that mediates information bidirectionally between patients and nurses. Patients require an accessible interface for on-demand explanations and daily guidance, while nurses need compact, interpretable summaries of how patients engage with their care. To address this gap, we propose Suzume-chan, a small plush "Pet-as-a-Friend" agent that serves as an embodied information hub. Suzume-chan is a soft, hand-sized object that supports speech-based interaction without requiring device operation or digital literacy (Fig. 2). Technically, it integrates local speech recognition with a standalone language model and retrieval-augmented generation (RAG) running on a nearby host.

This paper makes three contributions. First, we present Suzume-chan as a fully local ASR–LLM–RAG plush agent that provides patients with accessible, on-demand spoken explanations. Second, we report insights from a proof-of-concept workshop on how embodiment influences perceptions of latency, approachability, and trust in local LLM systems. Third, we identify vibrotactile feedback as a promising design direction for supporting interaction during processing delays, informing future development of expressive, bedside-friendly agents.

2 Study: Understanding Needs for Embodied Mediation

In nursing and medical care, patients and nurses often lack a shared understanding of care-related information. Patients receive explanations about routines, procedures, and self-management at fixed moments and in demanding formats, while nurses must infer patients' daily coping from brief interactions under time pressure, making fine-grained situational awareness difficult without added documentation burden.

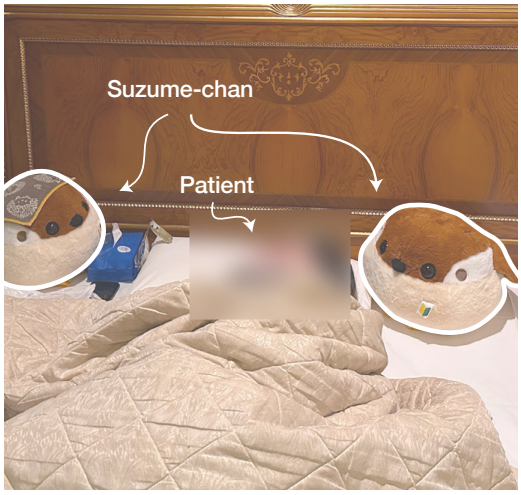

Figure 2: Example of Suzume-chan at the Bedside as a plush agent

While prior therapeutic robots such as Paro [6] show that soft, zoomorphic artifacts can reduce psychological barriers, they rarely function as part of an information infrastructure. Our goal therefore differs: to explore whether a small plush agent can act as a bidirectional mediator, offering approachable guidance for patients and compact behavioral cues for nurses.

To examine this, we conducted a proof-of-concept workshop using an early static prototype of Suzume-chan—a hand-sized "Pet-as-a-Friend" plush running on a local LLM/RAG backend. Qualitative analysis of 27 feedback segments identified three themes: Embodiment/Social Presence (T1; 15), System Performance/Functionality (T3; 11), and Trust/Privacy (T2; 1) as shown in figure 3.

A key finding was that embodiment strongly shaped perceptions of latency and trust. Participants described the plush as approachable and psychologically safe, yet noted that its static body made processing delays more salient. Many expected subtle movements or vibrations to signal intermediate states such as "listening" or "thinking," suggesting that physical expressiveness is a functional requirement rather than a purely aesthetic choice.

These insights inform three design requirements: (1) a fully local, privacy-preserving speech-based LLM–RAG pipeline; (2) low-latency conversational responses; and (3) embodied cues that mitigate unavoidable inference delays. The following section describes how the current system addresses (1) and (2), and how future work targets (3).

3 Implementation: Local Speech–LLM–RAG Architecture

Suzume-chan is an embodied audio interface connected to a nearby computational backend. The plush integrates a wireless microphone and speaker, enabling hands-free, device-free interaction suitable for patients with limited mobility (Fig. 1(a)). All conversational processing remains within the local network.

Speech input is transcribed using Apple's on-device Speech framework and passed to a backend combining an open-source large language model (gpt-oss-120B [1]) with retrieval-augmented

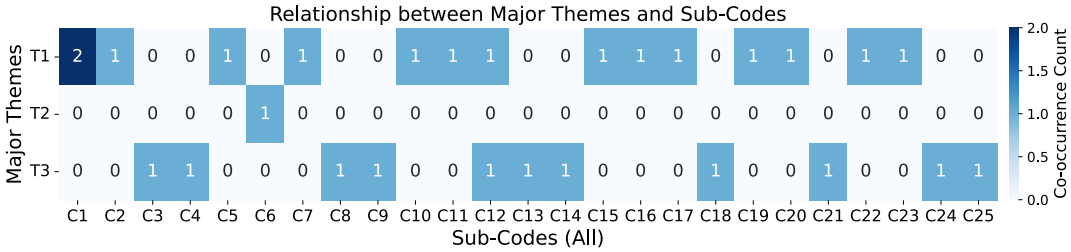

Figure 3: This matrix shows the co-occurrence relationships between major themes (T1–T3) and all sub-codes (C1–C25) identified in the qualitative analysis

Figure 4: A demonstration of Suzume-chan at WISS

generation [3]. Institutional procedures and daily guidance are stored in a vector database and retrieved to produce grounded responses, which are rendered via a Style-Bert-VITS2 text-to-speech pipeline[1] [4, 5] and played through the plush.

While this architecture enables fully local operation and supports psychological safety, workshop feedback indicated that response latency remains perceptible. Because the plush body provides no cues during inference, pauses felt longer and disrupted engagement, motivating the expressive interaction mechanisms explored next.

4 Demonstration: Toward Expressive, Low-Latency Bedside Interaction

The Interaction Demo prototype introduces two improvements inspired by the workshop.

Latency Optimization. We refined the inference pipeline through: (1) caching of high-frequency retrieval candidates, (2) streamlined ASR–LLM handoff, and (3) reduced-overhead text-to-speech synthesis. Together these reduce total response time and stabilize conversational pacing for bedside use.

Vibrotactile Embodiment. To address the strongest feedback theme, we integrate a small centrifugal actuator that provides controlled vibrotactile feedback during intermediate system states. These cues—gentle pulses when listening, a soft continuous vibration when "thinking," and a brief confirmation nudge before speaking—aim to convert silent processing intervals into meaningful, reassuring interaction moments. Because patients may interpret silence as uncertainty or hesitation, such embodied cues may help maintain trust and reduce repeated questioning.

The demo therefore presents Suzume-chan not only as a local LLM interface but as a dynamic mediator capable of signaling presence, activity, and attentiveness through non-verbal channels. We will present examples of latency-aware haptic behaviors and invite participants to evaluate how such expressiveness shapes immediacy, trust, and perceived responsiveness (Fig. 4).

5 Conclusion

By synthesizing early workshop insights with a privacy-preserving technical foundation, Suzume-chan advances toward an embodied information hub suitable for everyday inpatient life. The Interaction Demo highlights two directions essential for clinical viability: improving latency and enriching the plush with vibrotactile expressiveness. Together, these developments aim to transform a static bedside listener into an active, trustworthy mediator that supports both patients' information needs and nurses' situational awareness.

Acknowledgments

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